Solving Right Triangles (TE4)

Math 172 lecture notes · Section 4.3.1

Learning objective: use trigonometric functions and the Pythagorean Theorem to solve right triangles.

Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes Text from the handout Extra note (not in your handout)

Overview

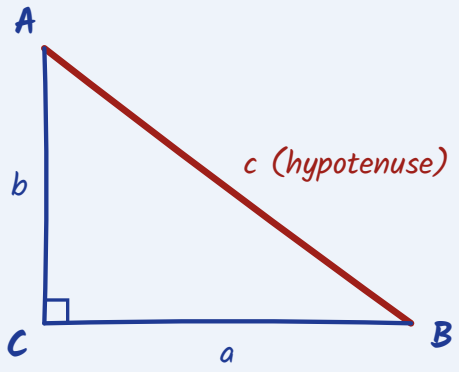
FROM THE HANDOUT

When asked to "solve" a triangle, this means to find all of the side lengths and angle measures within the triangle.

Note

FROM THE HANDOUT

When identifying the sides and angles of a right triangle, we typically use the convention where lowercase **a** and **b** are the legs, and **c** is the hypotenuse. Then we use the uppercase letters for angle measures. Each angle corresponds to the side opposite of the triangle.



Activity 1

Activity 1 · from your notes

Consider the right triangle $a = 35$, $b = 12$, and $c = 37$, with c being the hypotenuse of the triangle.

(a) Find $\cos A$.

Hint: Stand at angle A. Which side is opposite you? Which leg is next to you?

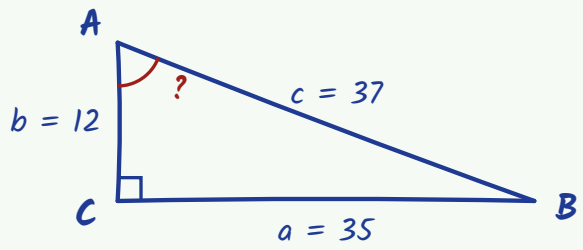
Answer (a)

First, check it really is a right triangle: $35^2 + 12^2 = 1225 + 144 = 1369 = 37^2$ ✓

From angle A: opposite = $a = 35$, adjacent = $b = 12$, hypotenuse = $c = 37$.

cos = adjacent over hypotenuse.

$$\cos A = \frac{12}{37}$$



(b) We can find angle A using sine, cosine, or tangent (all three sides are given). For each trig function, write the ratio that can be used to find the measure of angle A.

Answer (b)

- $\sin A = \frac{opp}{hyp} = \frac{35}{37}$
- $\cos A = \frac{adj}{hyp} = \frac{12}{37}$
- $\tan A = \frac{opp}{adj} = \frac{35}{12}$

(c) Now that we have set up a trigonometric ratio, how can we use these ratios to determine how big A is?

Hint: The ratio gives you a number. What "undoes" a function to get back its input?

Answer (c)

Each ratio gives the output of a trig function, but we want the input (the angle). To work backwards, **undo** the trig function with its **inverse**:

$$A = \sin^{-1}\left(\frac{35}{37}\right) = \cos^{-1}\left(\frac{12}{37}\right) = \tan^{-1}\left(\frac{35}{12}\right)$$

All three must give the same angle: same triangle, same angle.

Overview

FROM THE HANDOUT

Sometimes you will need to use inverse trigonometric functions (e.g. arcsin(x)) to find the measure of an angle. Recall from the previous example

that the inverse trig function keys on your calculator are usually denoted $\sin^{-1}$.

Inverse trig

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

$\sin^{-1}(x)$ means "the angle whose sine is x."

> ⚠ **Watch out:** $\sin^{-1}(x)$ is **not** $1 / \sin(x)$. The -1 means "undo," not "reciprocal."

Activity 2

Activity 2 · from your notes

Continuing from the previous activity...

(a) What is the trigonometric ratio for cos A?

Answer (a)

$$\cos A = \frac{12}{37}$$

(b) Use the inverse trig function $\cos^{-1}$ to find the measure of angle A. (Make sure your calculator is in degree mode!)

Answer (b)

$A = \cos^{-1}\left(\frac{12}{37}\right) \approx 71.075...°$

$$A \approx 71.08°$$

Check: $\sin^{-1}\left(\frac{35}{37}\right)$ and $\tan^{-1}\left(\frac{35}{12}\right)$ also give $\approx 71.08°$ ✓

(c) Now that we know the measure of angle A, find the measure of angle B.

> **Hint:** Do you need another trig function here, or is there a faster way?

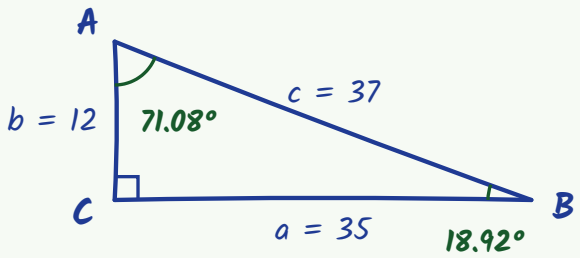
Answer (c)

No trig needed! $C = 90°$ and the angles add to $180°$, so $A + B = 90°$.

$B = 90° - 71.075...° \approx 18.92°$

$$B \approx 18.92°$$

Use the unrounded value of A when you subtract. Round only at the end.



Note

FROM THE HANDOUT

Determining all of the side lengths and angle measures of a right triangle is known as solving a right triangle. In previous examples, we were given all the sides of the triangle and used trigonometric ratios to determine the measure of the angles.

Game plan

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

1. Get the missing angle for free: the two small angles add to $90°$.
2. Pick a ratio: which side do I **know**, and which do I **want**, relative to the angle?
3. Finish with trig or the Pythagorean Theorem.
4. Check: $a^2 + b^2 = c^2$, and the hypotenuse is the longest side.

Activity 3

Activity 3 · from your notes

Solve the following triangles using your knowledge of right triangles, the Pythagorean Theorem and trigonometric functions. Be sure to draw a picture to help you determine the relationship between the angles and sides.

(a) In triangle ABC, $B = 53°$ and $c = 5$ meters (with c being the hypotenuse).

> **Hint:** Draw it first! Label the right angle and the hypotenuse, then mark what's given and what's missing.

Answer (a)

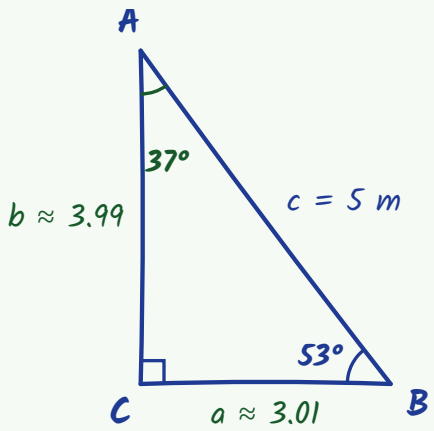
Angle: $A = 90° - 53° = 37°$

Side b is opposite B, so use sine: $\sin 53° = \frac{b}{5} \rightarrow b = 5 \sin 53° \approx 3.99$ m

Side a is adjacent to B, so use cosine: $\cos 53° = \frac{a}{5} \rightarrow a = 5 \cos 53° \approx 3.01$ m

Check: $3.993^2 + 3.009^2 \approx 25.00 = 5^2$ ✓

$A = 37°, B = 53°, C = 90°$; $a \approx 3.01$ m, $b \approx 3.99$ m, $c = 5$ m



(b) In triangle ABC, $A = 28°$ and $b = 29.3$ miles (with c being the hypotenuse).

> **Hint:** Draw it first! Label the right angle and the hypotenuse, then mark what's given and what's missing.

Answer (b)

Angle: $B = 90° - 28° = 62°$

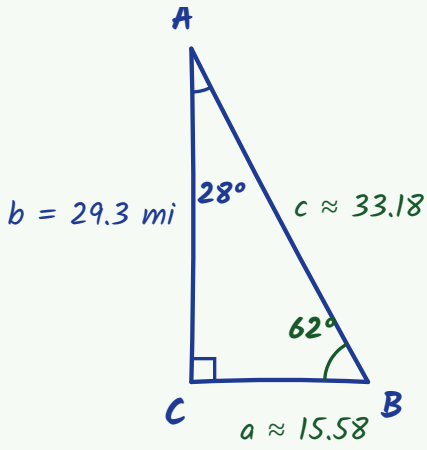
From A, b is adjacent. **Side a** is opposite, so use tangent: $\tan 28° = \frac{a}{29.3} \rightarrow a = 29.3 \tan 28° \approx 15.58$ mi

Side c is the hypotenuse, so use cosine: $\cos 28° = \frac{29.3}{c} \rightarrow c = \frac{29.3}{\cos 28°} \approx 33.18$ mi

> ⚠ **Watch out:** When the unknown is in the denominator, you **divide**.

Check: $\sqrt{(15.58^2 + 29.3^2)} \approx 33.18$ ✓

$A = 28°$, $B = 62°$, $C = 90°$; $a \approx 15.58$ mi, $b = 29.3$ mi, $c \approx 33.18$ mi



(c) In triangle ABC, $a = 8$ feet, $b = 17$ feet, and $c = 15$ feet (with b being the hypotenuse).

> **Hint:** Read carefully: which side is the hypotenuse this time? Where does that put the right angle?

Answer (c)

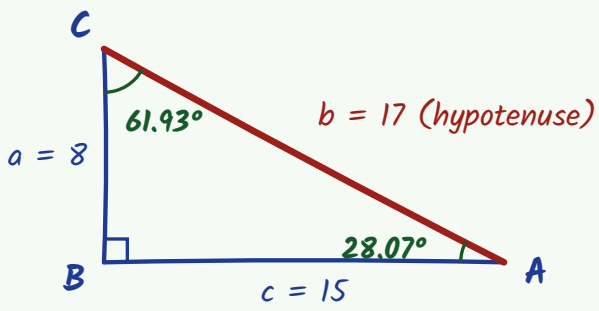
> ⚠ **Watch out:** The hypotenuse is **b**, so the right angle is **B**, not C.

Check: $8^2 + 15^2 = 64 + 225 = 289 = 17^2$ ✓ All sides are known, so we only need the angles.

Angle A is across from 8, with hypotenuse 17: $A = \sin^{-1}\left(\frac{8}{17}\right) \approx 28.07°$

Angle C $= 90° - 28.07...° \approx 61.93°$

$A \approx 28.07°$, $B = 90°$, $C \approx 61.93°$; $a = 8$ ft, $b = 17$ ft, $c = 15$ ft



Practice Example 4

Practice Example 4 · from your notes

A right triangle has a hypotenuse of length 10 and one angle of 35°. Find the length of the side opposite that angle, rounded to two decimal places.

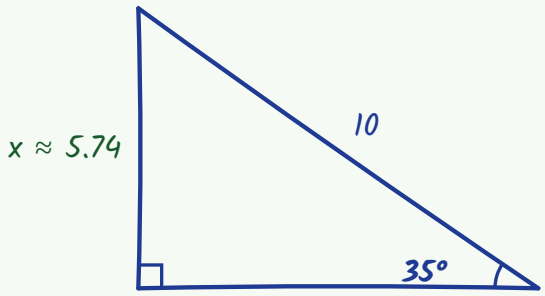
Try it, then check.

> **Hint:** Which ratio links opposite and hypotenuse?

Answer

Opposite and hypotenuse, so use sine: $\sin 35° = \frac{x}{10} \rightarrow x = 10 \sin 35° \approx 5.736\ldots$

$x \approx 5.74$



Practice Example 5

Practice Example 5 · from your notes

A right triangle has legs of length 7 and 24. Find the length of the hypotenuse and the measure of the angle opposite the side of length 7, rounded to one decimal place.

Try it, then check.

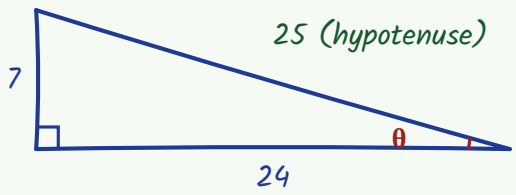
Answer

Hypotenuse: $c = \sqrt{(7^2 + 24^2)} = \sqrt{625} = 25$

Angle θ: 7 is opposite and 24 is adjacent, so $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$

hypotenuse = 25.0, $\theta \approx 16.3°$

"One decimal place" applies to both answers, so write 25.0, not 25.



Practice Example 6

Practice Example 6 · from your notes

A ladder leans against a wall, making a 70° angle with the ground. If the base of the ladder is 4 feet from the wall, how long is the ladder, rounded to two decimal places? Explain which trig ratio you chose and why.

Try it, then check.

> **Hint:** We know the adjacent side and want the hypotenuse. Which ratio connects those two?

Answer

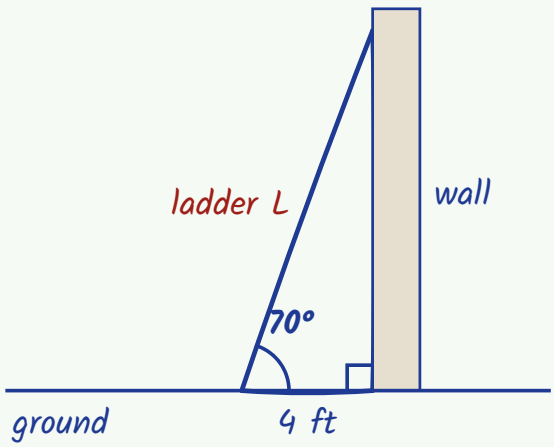
The ladder is the hypotenuse L. The 4 ft is adjacent to the 70° angle.

$$\cos 70° = \frac{4}{L} \rightarrow L = \frac{4}{\cos 70°} \approx 11.695\ldots$$

$L \approx 11.70$ ft

Why cosine? It's the only ratio that uses exactly the side we know (adjacent) and the side we want (hypotenuse). Sine would need the height up the wall, and tangent never uses the hypotenuse.

Check: a steep ladder should be much longer than its 4 ft base. 11.70 ft makes sense ✓



Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture.

Accessibility & help

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